Part 01 : Theoretical Questions

Q1 : What is the difference between static binding and dynamic binding? When does each one happen?

Q2 : What is the difference between method overloading and method overriding?

Q3 : What keywords are used for Method Overriding? What does each one mean ?

ROUTE

Part 02 : Practical (Extending the Movie Ticket Booking System)

In the previous assignments, you built a Movie Ticket Booking System with inheritance, properties, and static members. Now you will apply polymorphism to make the system flexible and extensible.

What you need to build :

1. Refactor the base Ticket class:

a. Add a PrintTicket() method that prints: TicketId, MovieName, Price, PriceAfterTax. Child classes should be able to provide their own version of this method.

b. Add two versions of a SetPrice method — one that takes a decimal (sets price directly) and one that takes a decimal base price and a decimal multiplier (sets price = base × multiplier).

2. In each child class, provide its own version of PrintTicket():

a. StandardTicket — prints the base ticket info and the SeatNumber.

b. VIPTicket — prints the base ticket info, LoungeAccess, and ServiceFee.

c. IMAXTicket — prints the base ticket info and whether it is 3D.

3. In the Cinema class, update PrintAllTickets() so it loops through the Ticket[] array and calls PrintTicket() on each one.

4. Create a static method ProcessTicket(Ticket t) that takes any Ticket and calls PrintTicket() on it.

5. In Main:

a. Create a Cinema and open it.

b. Create one StandardTicket, one VIPTicket, and one IMAXTicket.

c. Test both versions of SetPrice on one ticket.

d. Add all tickets to the Cinema and call PrintAllTickets().

e. Call ProcessTicket() with one of the tickets.

f. Close the Cinema.

Expected Output (Example) :

```
========== Cinema Opened ==========
Projector started.

========== SetPrice Test ==========
Setting price directly: 150
Setting price with multiplier: 100 x 1.5 = 150

========== All Tickets ==========
Ticket #1 | Inception | Price: 150 EGP | After Tax: 171.00 EGP
  Seat: A-5
Ticket #2 | Avengers | Price: 200 EGP | After Tax: 228.00 EGP
  Lounge: Yes | Service Fee: 50 EGP
Ticket #3 | Dune | Price: 180 EGP | After Tax: 205.20 EGP
  IMAX 3D: No

========== Process Single Ticket ==========
Ticket #2 | Avengers | Price: 200 EGP | After Tax: 228.00 EGP
```